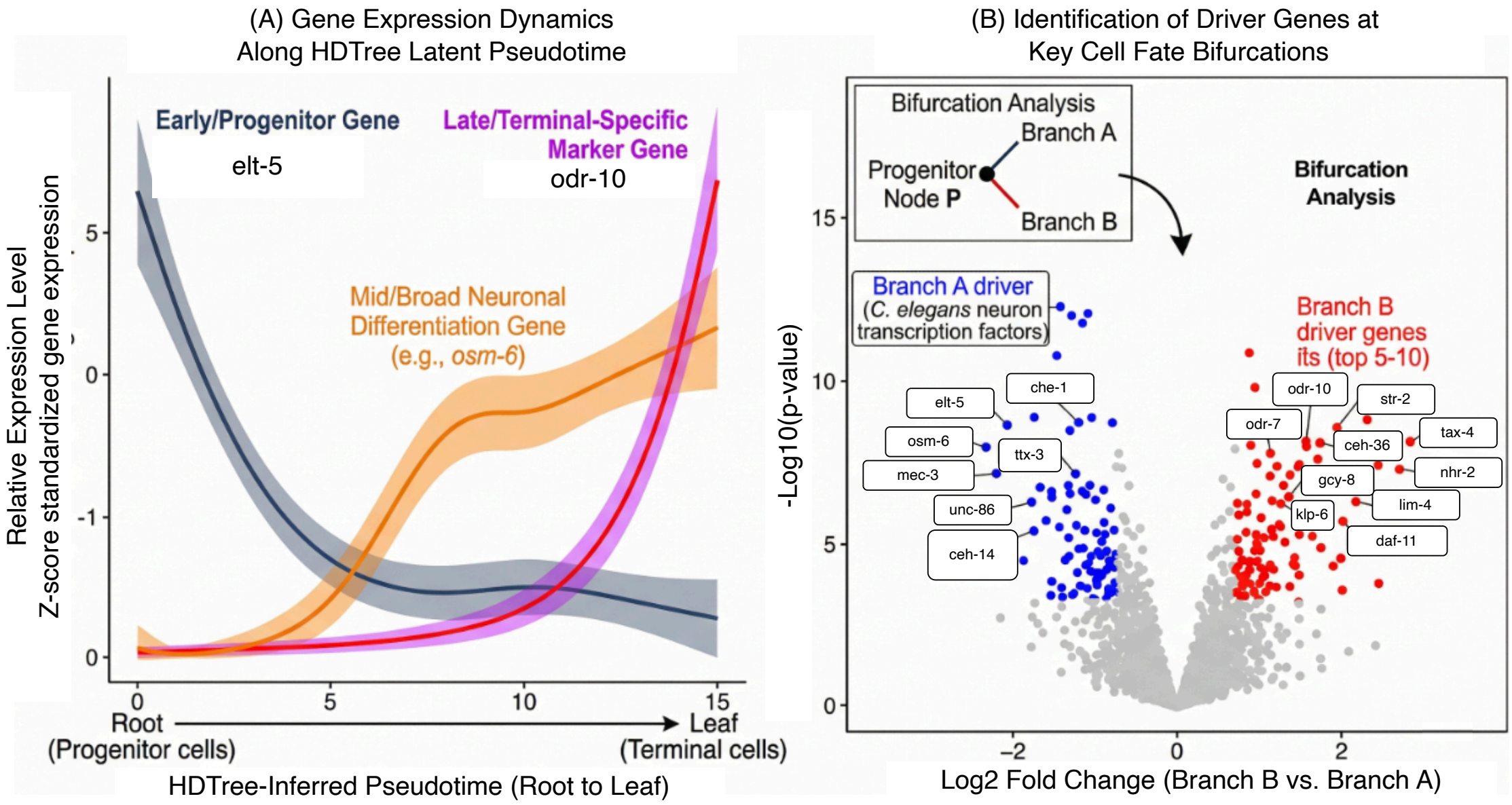
(A) Gene Expression Dynamics
Along HDTree Latent Pseudotime
Early/Progenitor Gene
elt-5
Late/Terminal-Specific
Marker Gene
odr-10
Mid/Broad Neuronal
Differentiation Gene
(e.g., osm-6)
Relative Expression Level
Z-score standardized gene expression
5
0
-1
0
0
5
10
15
Root
(Progenitor cells)
Leaf
(Terminal cells)
HDTree-Inferred Pseudotime (Root to Leaf)
(B) Identification of Driver Genes at
Key Cell Fate Bifurcations
Bifurcation Analysis
Branch A
Progenitor
Node P
Branch B
Bifurcation
Analysis
Branch A driver
(C. elegans neuron
transcription factors)
Branch B
driver genes
its (top 5-10)
elt-5
che-1
osm-6
ttx-3
mec-3
unc-86
ceh-14
odr-10
str-2
odr-7
tax-4
ceh-36
nhr-2
gcy-8
lim-4
klp-6
daf-11
-Log10(p-value)
15
10
5
0
−2
0
2
Log2 Fold Change (Branch B vs. Branch A)